# Topic 4: Data Link Layer, MAC Sub-layer & LAN Devices

Course outline weeks 3-4: medium access sub-layer, channel allocation, LAN protocols, ALOHA, IEEE standards, elementary data link protocols, sliding window, error handling + data link layer switching, bridges & routers.

## How often this topic is examined

| Question asked | Paper(s) | Marks |
|---|---|---|
| CSMA/CD (define / explain with diagram / in Ethernet) | 2018 Q3b, 2023 Q4a, SE2024 Q2b, ISA2024 Q4a | 3-5 |
| CSMA/CA; CSMA/CD vs CSMA/CA table | 2018 Q3b, 2023 Q4a | 3-5 |
| LLC and MAC sub-layers | 2023 Q5a [6], ISA2024 Q1c [6] | 6 |
| Error types (single-bit, burst) + detection methods; 3 error-detection strategies | 2023 Q4b [4], SE2024 Q3a [3] | 3-4 |
| FEC, Hamming distance, piggybacking | 2018 Q1b | 9 |
| Stop-and-wait protocol | 2018 Q4b | 4 |
| Pure ALOHA and slotted ALOHA (diagram) | ISA2024 Q4c | 8 |
| Structure of a MAC address | 2018 Q4d | 5 |
| ARP | 2018 Q3b.iii | 3 |
| Corrupted destination address on a bus LAN | 2025 Q5b | 6 |
| Bridge function in data-link switching | 2025 Q4c | 7 |
| Router vs switch (layers & functionality) | 2025 Q5d | 10 |
| Hub vs switch; repeater vs bridge; why a switch is intelligent | SE2024 Q4f [5], Q1b.ii [2], 2018 Q5d [3] | 2-5 |
| VLAN; network segmentation | 2018 Q5c [4], ISA2024 Q1b [3] | 3-4 |

**EXAM TIP:** CSMA/CD is in 4 of 5 papers - learn the flowchart. LLC/MAC is in 2 papers at 6 marks each.

## 1. The data link layer and its two sub-layers (IEEE 802)

**EXAM QUESTION** 2023 Q5a [3+3], ISA2024 Q1c [6] **[6]**
The data link layer in the IEEE standard is divided into two sub-layers. Discuss the functions performed by: i) LLC ii) MAC.



Made by King Tine, the AI Engineer

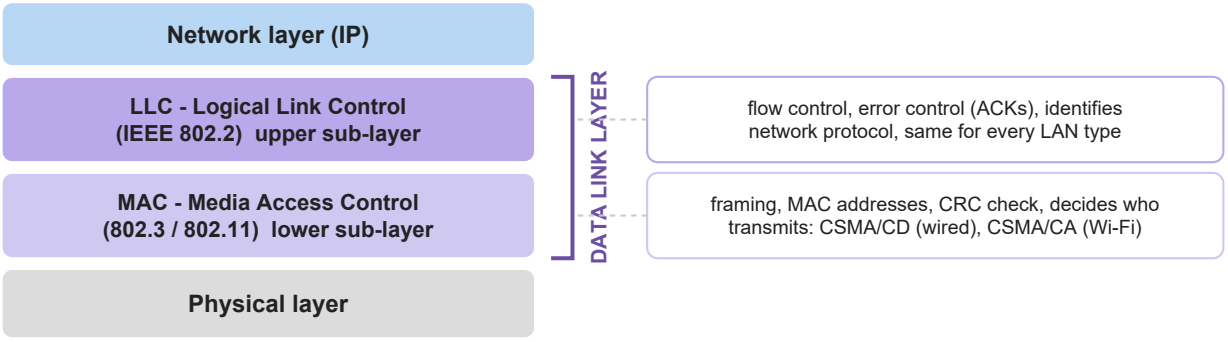



*Figure: the two sub-layers of the data link layer*

### LLC - Logical Link Control (upper)

- Interface with the **network layer** - identifies which network-layer protocol (IPv4, IPv6...) is carried, using **SAPs** (service access points) - allows multiplexing of protocols.
- **Flow control** - controls the rate at which data is sent (e.g. sliding window).
- **Error control** - acknowledgements and retransmission of lost/damaged frames (in connection-oriented LLC).
- Provides connectionless or connection-oriented services; it is **independent of the physical medium**, so the same LLC works over Ethernet, Wi-Fi, Token Ring.

### MAC - Media Access Control (lower)

- **Framing** - builds frames (preamble, destination & source MAC, type/length, data, FCS).
- **Physical (MAC) addressing** - adds 48-bit source/destination hardware addresses; recognises frames addressed to the station.
- **Media access control** - decides who may transmit on a shared medium and when: **CSMA/CD** (Ethernet 802.3), **CSMA/CA** (Wi-Fi 802.11), token passing.
- **Error detection** - computes/checks the **CRC (FCS)** and discards damaged frames.
- Interfaces with the physical layer; specific to each LAN technology.

**Data link layer overall functions:** framing, physical addressing, flow control, error control, access control - **node-to-node (hop-to-hop) delivery**.

## 2. Media access: CSMA/CD

> **EXAM QUESTION** ISA2024 Q4a [5], SE2024 Q2b [5], 2018 Q3b.i [3] **[3-5]**
> With the aid of a supporting diagram, explain the concept of CSMA/CD used by Ethernet. / In the description of the Ethernet architecture, explain the function and operation of CSMA/CD.

**CSMA/CD = Carrier Sense Multiple Access with Collision Detection.** Used by classic (half-duplex, shared/bus/hub) **Ethernet IEEE 802.3** so that many stations can share one cable.

- **Carrier Sense** - a station listens to the medium before sending ('listen before talk').
- **Multiple Access** - many stations share the same medium; any can transmit when it is free.
- **Collision Detection** - while transmitting, the station keeps listening ('listen while talk'); if the signal it sees differs from what it sent, a collision has happened.

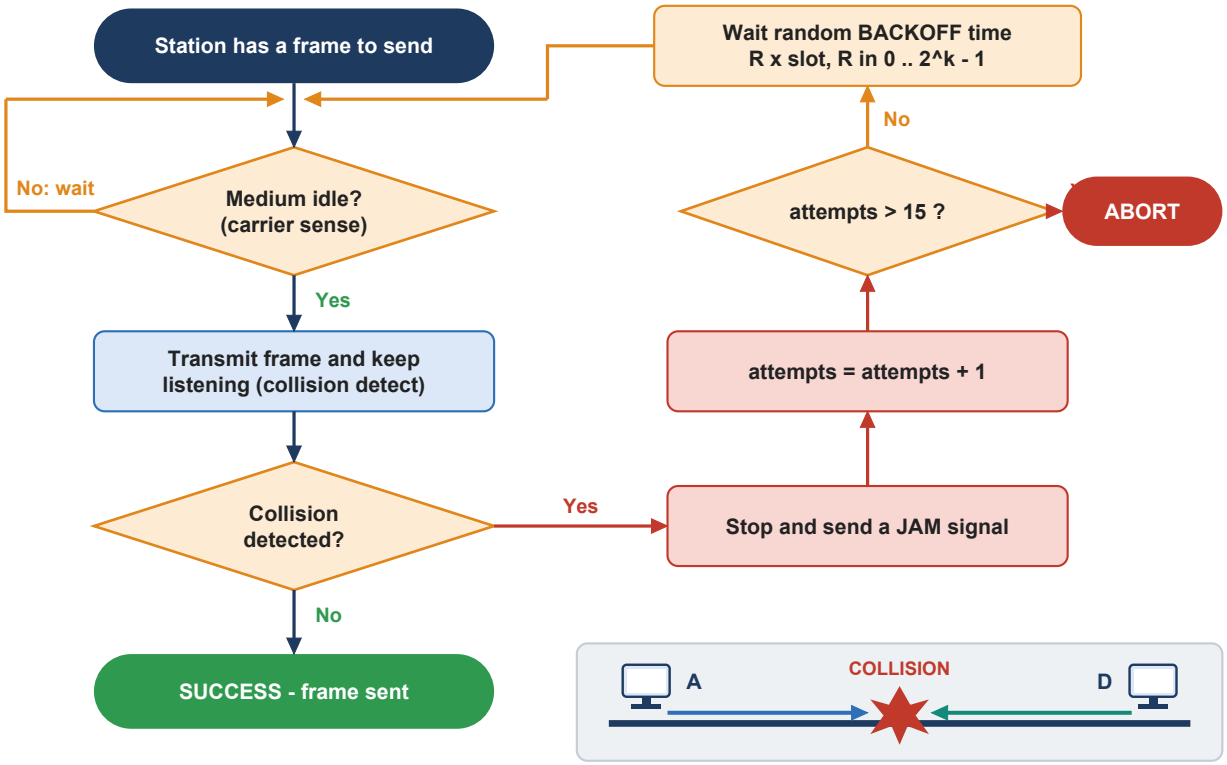



*Figure: CSMA/CD flowchart (and what a collision looks like)*

Important: a frame must be long enough (minimum 64 bytes in Ethernet) that the sender is still transmitting when the collision news returns (frame transmission time >= 2 x propagation delay). In modern **full-duplex switched Ethernet** there are no collisions, so CSMA/CD is not needed.

## 3. CSMA/CA and CSMA/CD vs CSMA/CA

**CSMA/CA = Carrier Sense Multiple Access with Collision Avoidance**, used in **wireless LANs (IEEE 802.11 Wi-Fi)**. Wireless stations cannot detect collisions reliably (they cannot send and listen at once, and the *hidden station* problem exists), so they try to **avoid** them:

1. Sense the channel; if idle wait an interframe space (**DIFS**).
2. Wait a random **backoff** (contention window) counting down only while idle.
3. Optionally send **RTS (Request To Send)**; the access point replies **CTS (Clear To Send)** - all other stations defer (NAV).
4. Send the data frame; receiver waits **SIFS** and returns an **ACK**. No ACK = assume collision -> retransmit with bigger backoff.

> **EXAM QUESTION** 2023 Q4a **[5]**
> Using a table, differentiate CSMA/CD from CSMA/CA.

| CSMA/CD | CSMA/CA |
|---|---|
| Collision **Detection** - deals with collisions **after** they happen | Collision **Avoidance** - tries to prevent collisions **before** they happen |
| Used in **wired** Ethernet LANs (IEEE 802.3) | Used in **wireless** LANs (IEEE 802.11 Wi-Fi) |
| Station transmits and listens at the same time | Wireless stations can't listen while sending; uses waiting times (IFS) & backoff |

| CSMA/CD | CSMA/CA |
|---|---|
| Jam signal + binary exponential backoff on collision | RTS/CTS handshake and ACK for every frame |
| No acknowledgement at MAC layer | Positive ACK required for each frame |
| Efficient on wired, less overhead | More overhead (IFS, RTS/CTS, ACK), lower efficiency |

## 4. ALOHA

**EXAM QUESTION** ISA2024 Q4c **[4 + 4]**
In the Aloha system, the basic idea is applicable to any system in which uncoordinated users are competing for a single shared channel. Explain using a diagram the functionality of: i) Pure ALOHA ii) Slotted ALOHA.

ALOHA was developed at the **University of Hawaii (1970)** for radio packet networks. It is a **random access** protocol.

### Pure ALOHA

- A station **sends whenever it has a frame** - no carrier sensing.
- It waits for an **ACK**; if no ACK arrives within a timeout (2 x max propagation time), it assumes a collision, waits a **random backoff** time and resends.
- A frame collides if any other frame starts within one frame time before or after it - **vulnerable time = 2 x Tfr**.
- **Maximum throughput = 1/(2e) = 18.4%** (at G = 0.5).

### Slotted ALOHA

- Time is divided into **slots** equal to one frame time (Tfr); stations are synchronised by a clock.
- A station may only start transmitting **at the beginning of a slot**.
- Collisions happen only if two stations pick the same slot, so **vulnerable time = Tfr** (half of pure ALOHA).
- **Maximum throughput = 1/e = 36.8%** (at G = 1) - double pure ALOHA. Needs synchronisation.

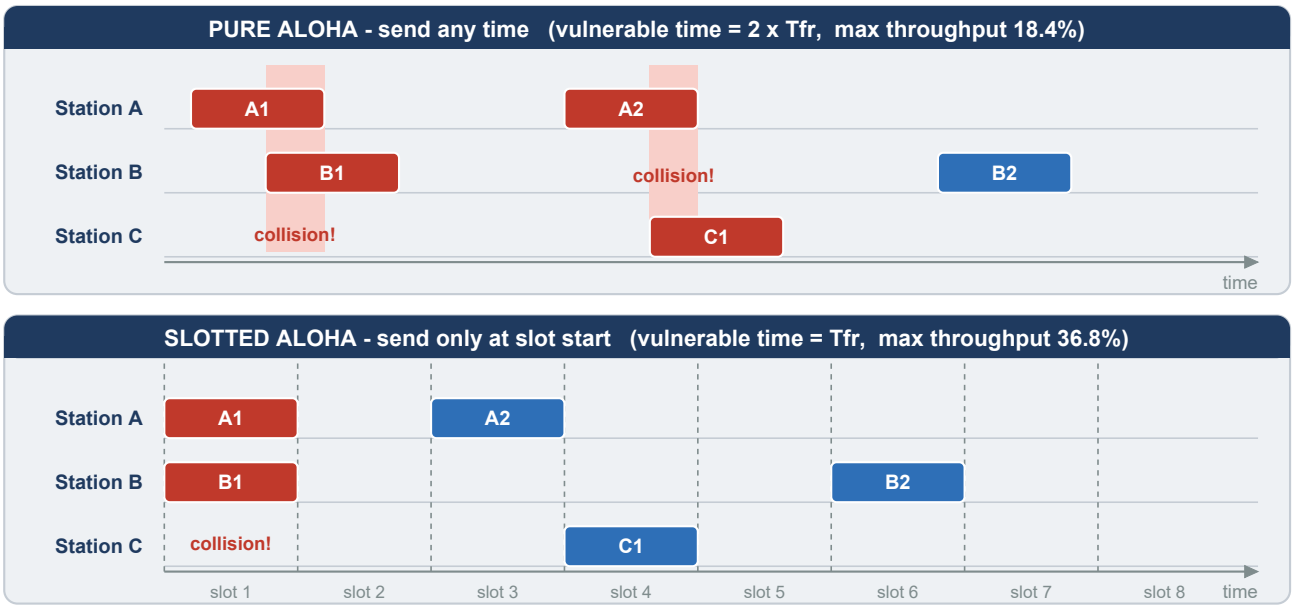



*Figure: pure ALOHA vs slotted ALOHA*

Related channel-allocation schemes: **static** (FDM, TDM) vs **dynamic** (ALOHA, CSMA, token passing). Controlled access: reservation, **polling**, **token passing** (Token Ring 802.5, **FDDI** - Fibre Distributed Data Interface: 100 Mbps dual counter-rotating fibre token ring, up to 200 km, the second ring gives fault tolerance).

## 5. Error detection and correction

**EXAM QUESTION** 2023 Q4b **[4]**
Copy and complete the table (Type of error | Description | Detection method).

| Type of error | Description | Detection method |
|---|---|---|
| Single-bit error | Only **one bit** of a data unit (byte/frame) changes from 0 to 1 or 1 to 0. e.g. 00000010 -> 00001010 | **Parity check** (VRC - simple even/odd parity) - detects all single-bit errors; also CRC, checksum |
| Burst error | **Two or more bits** in the data unit change (not necessarily consecutive); the burst length is from the first to the last corrupted bit. More common, caused by noise lasting longer than one bit time | **CRC (Cyclic Redundancy Check)** - detects all bursts shorter than the CRC length; also 2-D parity (LRC), checksum |

**EXAM QUESTION** SE2024 Q3a **[3]**
Explain any three error detection strategies used to control data transmission.

- **Parity check (VRC)** - one extra bit added so the total number of 1s is even (even parity) or odd. Receiver recounts; a mismatch shows an error. Detects single-bit and odd-numbered errors, misses even numbers of errors. **2-D parity (LRC)** adds a parity row too, catching more bursts.
- **CRC (Cyclic Redundancy Check)** - the sender divides the data (with n zeros appended) by an agreed generator polynomial using modulo-2 (XOR) division; the n-bit remainder (CRC) is appended. The receiver divides the whole frame by the same divisor: remainder 0 = no error. Used in Ethernet FCS. Very powerful.
- **Checksum** - data split into k-bit sections, added using 1's complement arithmetic; the complement of the sum is sent. Receiver adds all sections + checksum: result all 1s (complement = 0) = OK. Used in IP, TCP, UDP headers.

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CRC worked example: data 1001, generator 1011 (n = 3 -> append 000)
    1001000 / 1011 (modulo-2 / XOR division):
      1001 XOR 1011 = 010 -> bring down 0 -> 0100 (top bit 0: XOR 0000) = 100
      -> bring down 0 -> 1000 XOR 1011 = 011 -> bring down 0 -> 0110 (top bit 0) = 110
    remainder = 110  ->  transmitted codeword = 1001 110
    Receiver: 1001110 / 1011 -> remainder 000 = no error
Even-parity example: 1100001 has three 1s -> parity bit 1 -> sent 11000011
```

### Key terms (2018 Q1b - 3 marks each)

- **Forward Error Correction (FEC)** - the sender adds enough **redundant bits** that the **receiver can detect AND correct errors itself without asking for retransmission**. Used when retransmission is impossible or too slow: real-time audio/video, satellite links, deep-space, simplex links. Example: **Hamming code**, Reed-Solomon. (Opposite: **ARQ** - Automatic Repeat reQuest - detect and retransmit.)
- **Hamming distance** - the number of bit positions in which two codewords of equal length **differ**. Found by XORing them and counting the 1s: d(10101, 11110) -> 01011 -> **3**. The **minimum Hamming distance** (d_min) of a code determines power: to **detect** s errors d_min = s + 1; to **correct** t errors d_min = 2t + 1.

- **Piggybacking** - in bidirectional (full duplex) communication, the receiver does not send a separate ACK frame; instead it **attaches the acknowledgement to its next outgoing data frame** (ack field in the header). Improves efficiency/bandwidth use; disadvantage: if there is no data to send, the ACK is delayed (a timer forces a separate ACK).

## 6. Flow control: stop-and-wait & sliding window

**EXAM QUESTION** 2018 Q4b **[4]**
Explain the working principle of the stop and wait protocol.

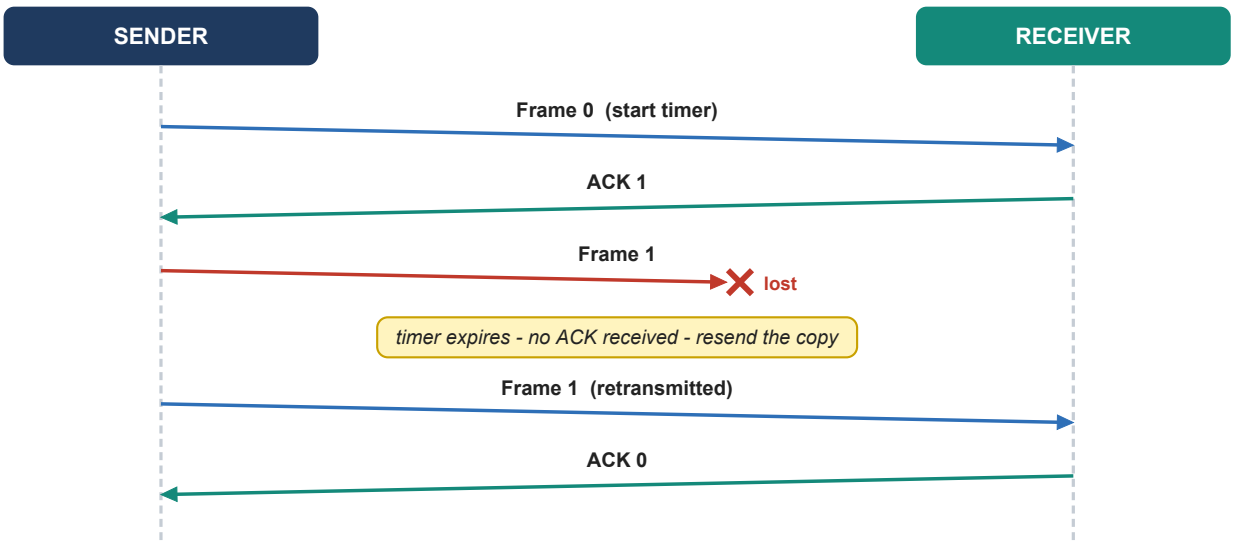



*Figure: stop-and-wait ARQ with a lost frame*

- Sender transmits **one frame** and keeps a copy, starts a **timer**, then **stops and waits** for an acknowledgement.
- The receiver checks the frame; if correct it sends an **ACK**; only then does the sender send the next frame.
- If the timer expires (frame or ACK lost/damaged), the sender **retransmits** the frame (Stop-and-Wait **ARQ**).
- Frames and ACKs are numbered alternately **0 and 1** (modulo-2 sequence numbers) so duplicates can be detected.
- Simple, but **inefficient** on long/fast links because the line is idle while waiting (window size = 1).

**Sliding window protocols** (outline topic): the sender may send **several frames (up to window size W)** before needing an ACK. **Go-Back-N ARQ**: sender window up to 2^m - 1, receiver window 1; on error the sender resends the damaged frame and **all frames after it**. **Selective Repeat ARQ**: windows up to 2^(m-1); only the damaged/lost frame is resent; receiver buffers out-of-order frames. Much better link utilisation than stop-and-wait.

## 7. MAC address structure

**EXAM QUESTION** 2018 Q4d **[5]**
With the aid of an example, describe the structure of a MAC address.

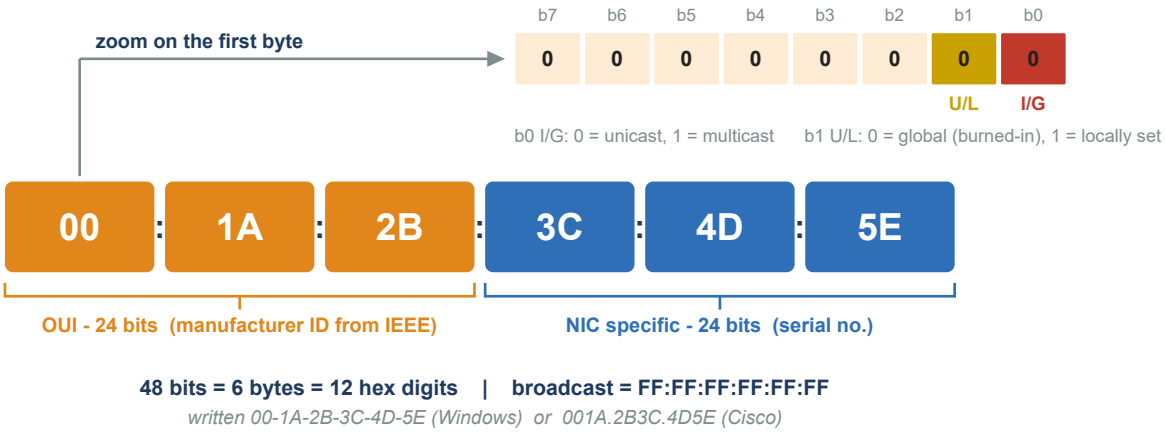



*Figure: structure of a 48-bit MAC address*

Written as 00-1A-2B-3C-4D-5E (Windows), 00:1A:2B:3C:4D:5E (Linux) or 001A.2B3C.4D5E (Cisco). Burned into the NIC ROM, used for delivery within a LAN; view with `ipconfig /all`.

## 8. ARP

**ARP (Address Resolution Protocol)** maps a known **IP address (Layer 3) to a MAC address (Layer 2)** on a local network. Host A wants to send to 192.168.1.20 but needs its MAC: (1) checks its **ARP cache**; (2) if absent, **broadcasts** an ARP Request to FF:FF:FF:FF:FF:FF: 'Who has 192.168.1.20? Tell 192.168.1.10'; (3) the owner replies with a **unicast ARP Reply** containing its MAC; (4) A stores it in the cache and sends the frame. **RARP** does the reverse (MAC -> IP).

## 9. Corrupted destination address on a bus

> **EXAM QUESTION** 2025 Q5b **[3 + 3]**
> Suppose a computer sends a frame to another computer on a bus topology LAN. The physical destination address of the frame is corrupted during transmission: i) Explain what happens to the frame. ii) Explain how the sender can be informed about the situation.

**i)** On a bus every station receives the frame. Each station compares the destination MAC address with its own. Because the address is corrupted, the **intended receiver will not recognise the frame** as its own and ignores it. Also, the corruption changes the bits, so the **CRC/FCS check fails** at every station and the frame is **discarded** (even if the corrupted address happened to match another station). The frame is effectively lost; it eventually reaches the terminators and is absorbed.

**ii)** The data link layer of Ethernet does not send error messages for a discarded frame, so the sender is informed **indirectly** through the **lack of an acknowledgement**: in protocols using ACKs (Stop-and-Wait / sliding window ARQ, or TCP at the transport layer) the sender's **timer expires** without an ACK, so it concludes the frame/segment was lost and **retransmits** it. (A NAK can be used if the receiver can recognise the frame.)

## 10. LAN devices: hub, repeater, bridge, switch, router

> **EXAM QUESTION** 2025 Q4c **[7]**
> Describe the function of a network bridge and how it operates in a Data Link Layer switching environment.

- A **bridge** is a **Layer 2** device that connects two (or more) LAN segments and **filters and forwards frames based on MAC addresses**. It divides the network into separate **collision domains** while remaining one broadcast domain.

- **Learning:** when a frame arrives, the bridge records the **source MAC** and the port it arrived on in its **MAC/forwarding table** (transparent/learning bridge - no configuration needed).
- **Filtering:** if the destination MAC is on the **same port** the frame came from, the bridge drops it (traffic stays local).
- **Forwarding:** if the destination is known on another port, the frame is sent **only to that port**.
- **Flooding:** if the destination is unknown, or it is a broadcast/multicast, the frame is sent out all ports except the incoming one.
- **Ageing:** table entries time out (~300 s) so moved hosts are relearned.
- **Spanning Tree Protocol (STP, IEEE 802.1D)** prevents loops when bridges are connected redundantly by blocking some ports.
- A **switch** is a multi-port bridge doing this in hardware (ASICs) at wire speed.

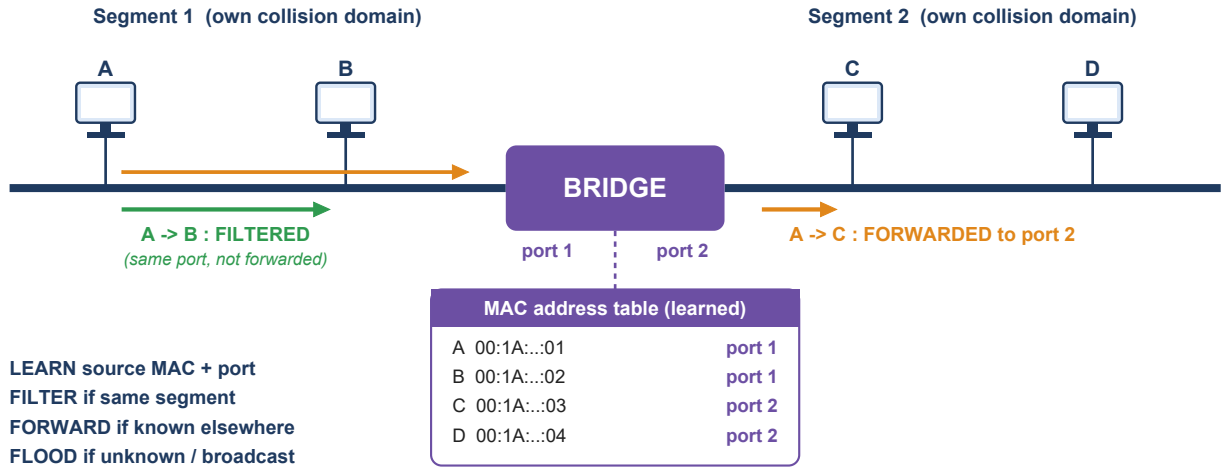



*Figure: a learning bridge filtering and forwarding frames*

**EXAM QUESTION** SE2024 Q4f **[5]**
Differentiate between hub and switch on the basis of layer, ports, device type, speed.

| Basis | Hub | Switch |
|---|---|---|
| OSI layer | Layer 1 - Physical | Layer 2 - Data Link (L3 switches also exist) |
| Ports | Typically 4/8/12; all ports share one collision domain | 24/48 ports; **each port is its own collision domain** |
| Device type | **Passive/dumb** - broadcasts every frame to all ports | **Intelligent/active** - uses a MAC table to send frames only to the destination port |
| Speed / duplex | Shared bandwidth (10/100 Mbps), **half duplex** | Dedicated bandwidth per port (100 Mbps - 10 Gbps), **full duplex** |
| Addressing | No addresses used | Uses MAC addresses |
| Security / collisions | Many collisions; anyone can sniff traffic | No collisions in full duplex; more secure |

**Why a switch is said to be intelligent (2018 Q5d, 3 marks):** it **learns** MAC addresses and builds a MAC address table, **forwards frames only to the port of the destination** (not to everyone like a hub), filters local traffic, checks frames for errors (store-and-forward), supports **VLANs**, full duplex and QoS, and (managed switches) can be configured and monitored.

**Repeater vs bridge (SE2024 Q1b.ii, 2 marks):** a **repeater** (Layer 1) simply **regenerates/amplifies** the signal bit by bit to extend the cable distance - it does not filter and forwards everything including collisions.

A **bridge** (Layer 2) reads MAC addresses and **filters/forwards frames** between segments, reducing traffic and separating collision domains.

> **EXAM QUESTION** 2025 Q5d **[10]**
> Discuss the purpose of a router and a switch in a network. Which OSI layers do they operate on, and what is the difference between them in terms of functionality?

| Point | Switch | Router |
|---|---|---|
| Purpose | Connects devices **within the same network (LAN)** | Connects **different networks** (LAN to LAN, LAN to WAN/Internet) |
| OSI layer | **Layer 2 - Data Link** (multilayer switch also L3) | **Layer 3 - Network** |
| Address used | **MAC** addresses, MAC address table | **IP** addresses, routing table |
| Data unit | Frames | Packets |
| Decision | Forwards frames to the correct port | Determines the **best path** using routing protocols (RIP, OSPF, EIGRP) |
| Broadcasts | Forwards broadcasts - one broadcast domain (unless VLANs) | **Blocks broadcasts** - each interface is a separate broadcast domain |
| Extra features | VLANs, port security, PoE | NAT, DHCP, firewall/ACLs, VPN, WAN interfaces |
| Example | Lab switch connecting 24 PCs | Home/office router connecting the LAN to the ISP |

## 11. VLANs & network segmentation

**VLAN (Virtual LAN)** (2018 Q5c, 4 marks) - a **logical grouping of devices into a separate broadcast domain on a switch, regardless of physical location** (configured in software, IEEE **802.1Q** tagging). **Uses:** separate departments (Accounts, HR, IT) on the same switches; **security** (sensitive traffic isolated); **reduce broadcast traffic** and improve performance; flexibility - move users without recabling; separate voice and data traffic (voice VLAN, QoS). Inter-VLAN traffic needs a router or Layer 3 switch.

**Network segmentation** (ISA2024 Q1b, 3 marks) - dividing a large network into smaller **segments/subnets** (with routers, switches, VLANs, subnetting or firewalls). **Benefits:** reduces congestion & broadcast/collision domains -> better **performance**; improves **security** (contains attacks/malware, limits lateral movement, applies access control between zones e.g. DMZ); easier **management** and troubleshooting.

## 12. IEEE 802 standards overview (outline)

| Standard | Technology |
|---|---|
| 802.1 | Bridging, VLANs (802.1Q), STP (802.1D) |
| 802.2 | LLC |
| 802.3 | Ethernet (CSMA/CD) |
| 802.5 | Token Ring |
| 802.11 | Wireless LAN / Wi-Fi (CSMA/CA) |
| 802.15 | WPAN - Bluetooth (802.15.1), Zigbee (802.15.4) |

| Standard | Technology |
|---|---|
| 802.16 | WiMAX (wireless MAN) |

**REMEMBER:** Detection: Parity, CRC, Checksum. Correction: Hamming code (FEC) or retransmit (ARQ: Stop-and-Wait, Go-Back-N, Selective Repeat). CD = wired, react after; CA = wireless, prevent before. Pure ALOHA 18.4%, Slotted 36.8%.